

Error for Different Shapes

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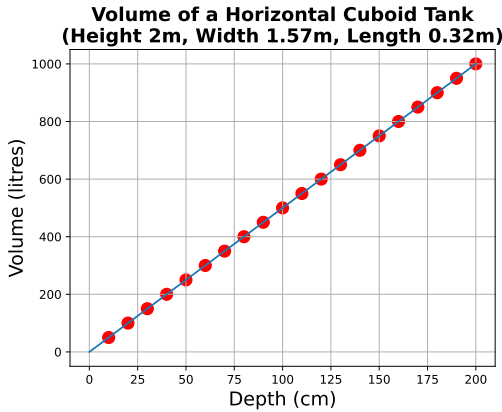
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It is important to note that a heating oil tank does not just come in the shape of a cylinder. Many shapes are used such as a cube shape. Other shapes are better for storing different liquids too, like how a sphere is used to hold large quantities of water [1]. We want to investigate whether there is a difference between reading error rate in a cuboid and a sphere.

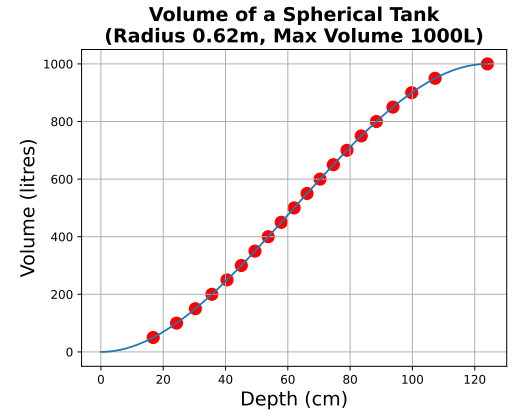
1 Dipstick Calibration

In Figure (1) we can see the dipstick calibration for a cuboid and a sphere. From Figure (1a) we can see that the dipstick calibration for a cuboid and it is linear and the marks on the dipstick are evenly shaped out. This is because a cuboid has the same cross section through out the shape so we would not expect any variation.

From Figure (1b) we can see the dipstick calibration for a sphere. We can see the marks are spaced out unevenly. This is because as the tank get empty and full the cross section of the sphere gets narrower.



(a) Cuboid

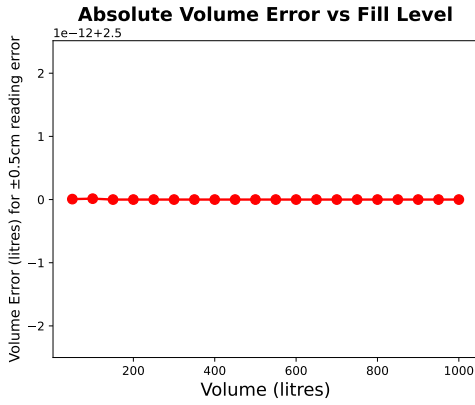


(b) Sphere

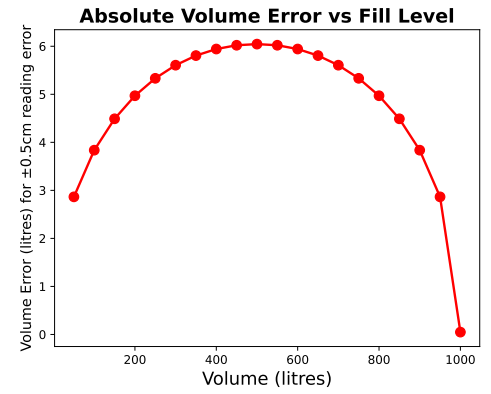
Figure 1: Dipstick Calibration Marks for a Cuboid and a Sphere

2 Absolute Error

In Figure (2) we can see the absolute error of reading the dipstick $\pm 0.5\text{cm}$ off. Figure (2a) shows the absolute error of a cuboid. As we can see the error is exactly the same throughout the shape and there is little to no error. From Figure (2b) we can see the absolute error of a sphere. As we would expect the error is the highest in the middle of the sphere where the liquid surface is the widest. Then the error is the lowest when the liquid surface is narrowest (at the bottom and top of the tank)



(a) Cuboid

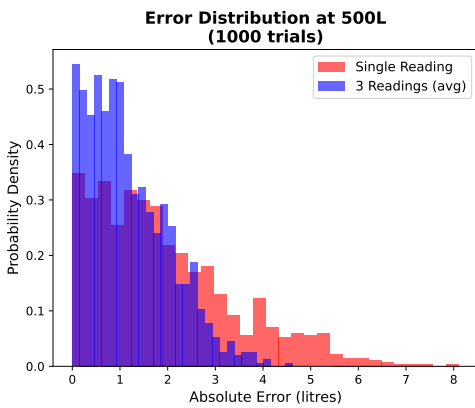


(b) Sphere

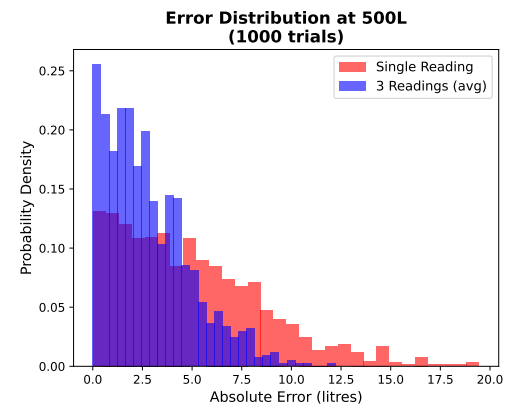
Figure 2: Absolute Error vs Fill Level of a Cuboid and a Sphere

3 Error Distribution for Multiple Trials

To make the readings more accurate we ran a Monte Carlo simulation where we read the dipstick 3 time for 1000 trials. From Figure (3) we can see the result of these trials for both the cuboid and the sphere. It is clear that the 3 readings (blue) is more accurate than the single reading (red) in both shapes.



(a) Graph 1



(b) Graph 2

Figure 3: Three graphs displayed side-by-side

4 Code

(see [2] for the code for the different shape error analysis)

5 References

References

- [1] Chemical Engineering World. (2019). Spherical Storage Tank Design. [online] (Accessed on 25/02/26)
- [2] Elliemat21 (2025). AMA3020SoloProject/DipstickDifferentShapes.ipynb at main · Elliemat21/AMA3020SoloProject. [online] GitHub. (Accessed on 25/02/26)